Math 54 — Week 7

Solutions

P1.

Idea. Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 = \mathbf{x}$.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 3 & 5 & 9 \end{array}\right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -1 & -3 \end{array}\right]$$

$c_2 = 3$, $c_1 = 4 - 6 = -2$. $[\mathbf{x}]_{\mathcal{B}} = (-2, 3)$. Check: $-2(1,3) + 3(2,5) = (4,9)$.

P2.

Idea. Work with coordinate vectors in the standard basis $\{1, t, t^2\}$.

1. (a) Solve $c_1(1+t) + c_2(t+t^2) + c_3(1+t^2) = 3 + 2t - t^2$. Matching coefficients: constant $c_1 + c_3 = 3$; t: $c_1 + c_2 = 2$; t^2: $c_2 + c_3 = -1$. Adding all three: $2(c_1 + c_2 + c_3) = 4$, so $c_1 + c_2 + c_3 = 2$. Then $c_2 = 2 - 3 = -1$, $c_3 = 2 - 2 = 0$, $c_1 = 3$. $[p]_{\mathcal{B}} = (3, -1, 0)$. Check: $3 + 3t - t - t^2 = 3 + 2t - t^2$.
2. (b) Coordinate vectors $(1, -1, 0)$, $(0, 1, -1)$, $(1, 0, -1)$. The third is the sum of the first two, so the polynomials are dependent: $1 - t^2 = (1 - t) + (t - t^2)$.

P3.

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 3 \\ 1 & 1 & 2 & 4 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 1 \end{array}\right], \qquad P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}.$$

$$P_{\mathcal{B}\leftarrow\mathcal{C}} = (P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1} = \frac{1}{1-3}\begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & 3/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

$$[\mathbf{x}]_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}(2, -1) = (2 - 3,\ 2 - 1) = (-1, 1).$$

Check: $\mathbf{x} = 2(1,2) - (3,4) = (-1, 0)$ and $-1(1,1) + 1(0,1) = (-1, 0)$.

P4. Multiply:

$$A\mathbf{v} = (-4 + 0 + 1,\ 2 + 1 + 0,\ 2 + 0 + 1) = (-3, 3, 3) = 3\mathbf{v},$$

so $\mathbf{v}$ is an eigenvector of A for the eigenvalue 3. For $\lambda = 4$:

$$A - 4I = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ -2 & 0 & -3 \end{bmatrix}, \qquad \det(A - 4I) = 0 \cdot (9 - 0) - 0 + 1 \cdot (0 - 6) = -6 \neq 0,$$

so $A - 4I$ is invertible and 4 is not an eigenvalue.

P5.

Idea. Column 2 of $A-\lambda I$ has a single nonzero entry; expand down it and keep factors.

$$\det(A-\lambda I) = (1-\lambda)\det\begin{bmatrix}4-\lambda & 1\\ -2 & 1-\lambda\end{bmatrix} = (1-\lambda)\big((4-\lambda)(1-\lambda)+2\big) = (1-\lambda)(\lambda^2-5\lambda+6) = (1-\lambda)(\lambda-2)(\lambda-3).$$

Eigenvalues $1,2,3$. For $\lambda=2$:

$$A-2I = \begin{bmatrix}2 & 0 & 1\\ -2 & -1 & 0\\ -2 & 0 & -1\end{bmatrix} \to \begin{bmatrix}1 & 0 & 1/2\\ 0 & 1 & -1\\ 0 & 0 & 0\end{bmatrix},$$

so $x_1 = -\frac{1}{2}x_3$, $x_2 = x_3$. Basis: $\{(-1,2,2)\}$.

P6. For a 2×2 matrix the characteristic polynomial is $\lambda^2 - (\text{trace})\lambda + \det$.

(a) For $\left[\begin{smallmatrix}2 & 7\\ 7 & 2\end{smallmatrix}\right]$:

$$\lambda^2 - 4\lambda + (4-49) = \lambda^2 - 4\lambda - 45 = (\lambda-9)(\lambda+5),$$

so the eigenvalues are 9 and -5.

(b) For $\left[\begin{smallmatrix}5 & -3\\ -4 & 3\end{smallmatrix}\right]$: $\lambda^2 - 8\lambda + (15-12) = \lambda^2 - 8\lambda + 3$, so the eigenvalues are $4 \pm \sqrt{13}$.

P7. The matrix is upper triangular: eigenvalues are the diagonal entries $5,3,5,1$. So 5 (multiplicity 2), 3 and 1 (multiplicity 1 each). The characteristic polynomial is $(5-\lambda)^2(3-\lambda)(1-\lambda)$.

P8. (a) Apply A to $A\mathbf{x} = \lambda\mathbf{x}$, and add $3\mathbf{x}$ to it:

$$A^2\mathbf{x} = A(\lambda\mathbf{x}) = \lambda A\mathbf{x} = \lambda^2\mathbf{x} \quad \text{and} \quad (A+3I)\mathbf{x} = \lambda\mathbf{x} + 3\mathbf{x} = (\lambda+3)\mathbf{x}.$$

Eigenvalues λ^2 and $\lambda+3$. (b) If $\lambda = 0$ then $A\mathbf{x} = \mathbf{0}$ with $\mathbf{x} \neq \mathbf{0}$, so A would not be invertible. Multiply $A\mathbf{x} = \lambda\mathbf{x}$ by A^{-1}: $\mathbf{x} = \lambda A^{-1}\mathbf{x}$, so $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.